

Numerical Aerodynamic Investigation and Stall Characterization of the NACA 2412 Airfoil Using ANSYS Fluent and XFLR5

Aerodynamics & CFD Technical Report

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Abstract

A two-dimensional numerical investigation of the aerodynamic characteristics of a cambered NACA 2412 airfoil was conducted across angles of attack ranging from $\alpha = 0^\circ$ to $\alpha = 15^\circ$ at a chord-based Reynolds number of $Re \approx 3.42 \times 10^5$ ($V_\infty = 50$ m/s). The computational model utilized a fully turbulent Reynolds-Averaged Navier-Stokes (RANS) formulation coupled with the Shear Stress Transport ($k-\omega$ SST) closure model in ANSYS Fluent. Inflow directionality was controlled via an upward-tilted domain topology. Numerical predictions for lift (C_L), drag (C_D), and surface pressure distributions (C_p) were validated against panel method and boundary-layer coupled solutions from XFLR5. Excellent agreement was observed in the linear lift regime ($\alpha = 0^\circ$ to 10°). At $\alpha = 15^\circ$, the onset of aerodynamic stall triggered large-scale boundary-layer separation and self-sustained, limit-cycle shedding oscillations, highlighting the physical transition from steady attached flow to an unsteady vortex-shedding wake regime.

1 Introduction and Objectives

The NACA 2412 profile is a four-digit asymmetric airfoil characterized by a 2% maximum camber located at 40% chord from the leading edge and a maximum thickness of 12% chord. The objective of this study is to:

1. Evaluate lift and drag coefficients (C_L, C_D) across $\alpha = 0^\circ, 5^\circ, 10^\circ, 15^\circ$ using RANS $k-\omega$ SST.
2. Validate non-dimensional surface pressure coefficient (C_p) distributions against XFLR5 (XFoil-based higher-order panel method).

3. Investigate near-wall boundary layer development, suction peak behavior, and stall dynamics at high incidence.

2 Numerical Methodology and Domain Setup

2.1 Governing Equations and Reference Parameters

The flow is modeled as incompressible and steady using the two-dimensional continuity and Reynolds-Averaged Navier-Stokes (RANS) equations:

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (1)$$

$$\rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \rho \overline{u'_i u'_j} \right] \quad (2)$$

Turbulent stresses are resolved via Menter's two-equation k - ω SST model. The operating parameters set within Fluent's Reference Values panel are summarized in Table 1.

Table 1: Computational Reference Values and Operating Conditions

Parameter	Symbol / Setting	Value
Chord Length	c	0.1 m (100 mm)
Unit Span	b	1.0 m
Reference Planform Area	$A_{\text{ref}} = c \cdot b$	0.1 m ²
Freestream Velocity	V_{∞}	50.0 m/s
Air Density	ρ	1.225 kg/m ³
Dynamic Viscosity	μ	1.7894×10^{-5} kg/(m · s)
Reynolds Number	$\text{Re} = \rho V_{\infty} c / \mu$	3.423×10^5
Mach Number	M_{∞}	0.146 (Incompressible regime)

2.2 Tilted Domain Boundary Formulation

Rather than re-meshing rotated airfoils, an upward-tilted farfield domain was implemented. The freestream enters normal to the boundary with an inclination angle α . In ANSYS Fluent, force monitors project total surface force vectors via direction dot products:

$$F_{\text{reported}} = \vec{F}_{\text{net}} \cdot \hat{d} = F_x d_x + F_y d_y \quad (3)$$

To ensure reported coefficients represent true wind-axis Aerodynamic Drag (parallel to freestream) and Aerodynamic Lift (perpendicular to freestream), report definition unit vectors were specified as:

$$\hat{d}_{\text{Drag}} = (\cos \alpha) \hat{i} + (\sin \alpha) \hat{j} \quad (4)$$

$$\hat{d}_{\text{Lift}} = (-\sin \alpha) \hat{i} + (\cos \alpha) \hat{j} \quad (5)$$

Table 2: Force Report Monitor Direction Vectors by Angle of Attack

α	$\cos \alpha$	$\sin \alpha$	$\hat{d}_{\text{Drag}} = (X, Y)$	$\hat{d}_{\text{Lift}} = (X, Y)$
0°	1.0000	0.0000	(1.0000, 0.0000)	(0.0000, 1.0000)
5°	0.9962	0.0872	(0.9962, 0.0872)	(-0.0872, 0.9962)
10°	0.9848	0.1736	(0.9848, 0.1736)	(-0.1736, 0.9848)
15°	0.9659	0.2588	(0.9659, 0.2588)	(-0.2588, 0.9659)

3 Results and Validation

3.1 Aerodynamic Force Coefficients Summary

The calculated lift and drag coefficients from ANSYS Fluent are compared with XFLR5 panel calculations in Table 3.

Table 3: Comparison of Aerodynamic Coefficients between Fluent and XFLR5

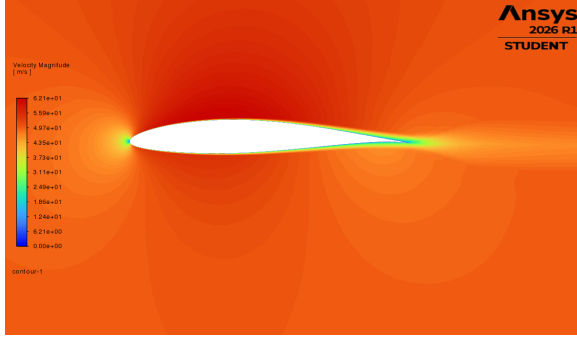
α	ANSYS Fluent ($k\text{-}\omega$ SST)		XFLR5 (XFoil Coupled)		Relative Offset	
	C_L	C_D	C_L	C_D	ΔC_L	C_D Ratio
0°	0.2827	0.01490	0.23075	0.00738	+22.5%	$2.02\times$
5°	0.8235	0.01970	0.80870	0.01023	+1.83%	$1.93\times$
10°	1.2287	0.03520	1.19539	0.02248	+2.79%	$1.57\times$
15°	[0.36, 1.54]*	[0.105, 0.350]*	1.28845	0.06362	(Stall)	(Unsteady)

*Denotes limit-cycle periodic oscillation bounds under steady-state solver formulation.

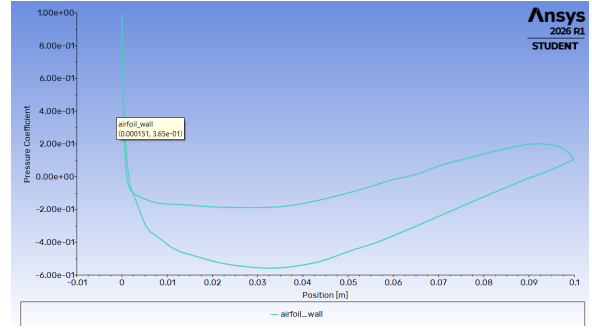
3.2 Analysis of Flow Characteristics by Angle of Attack

3.2.1 Low Incidence Regime ($\alpha = 0^\circ$)

At $\alpha = 0^\circ$, positive camber generates lift ($C_L = 0.2827$). The maximum local velocity reaches 62.1 m/s ($1.24 U_\infty$) at the upper suction crest, with a peak negative pressure coefficient $C_{p,\min} \approx -0.56$, matching XFLR5 ($C_{p,\min} = -0.5693$). Fluent reports a higher drag coefficient (0.01490 vs. 0.00738) due to the fully turbulent boundary layer assumption across the entire profile, whereas XFLR5 models laminar boundary layer run until natural transition occurs at $x/c = 0.765$ on the suction surface and $x/c = 0.916$ on the pressure surface.



(a) Velocity magnitude contour ($V_{\max} = 62.1 \text{ m/s}$)

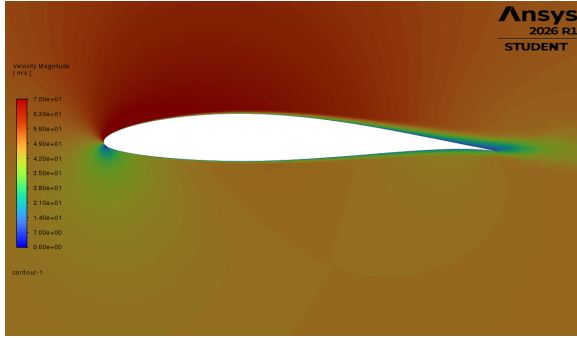


(b) Surface pressure distribution (C_p vs. x)

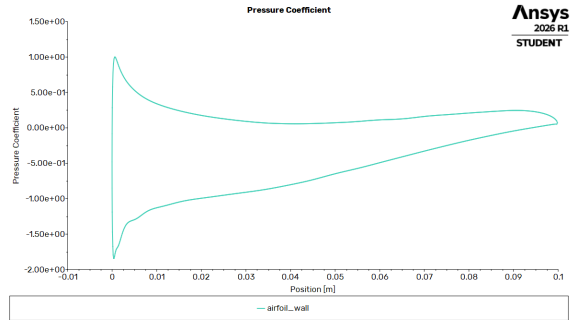
Figure 1: Aerodynamic flow field at $\alpha = 0^\circ$.

3.2.2 Linear Moderate Lift Regime ($\alpha = 5^\circ$)

At $\alpha = 5^\circ$, the airfoil operates within the linear lift region ($C_L = 0.8235$), agreeing within 1.83% of XFLR5 ($C_L = 0.8087$). The leading-edge suction peak intensifies to $C_{p,\min} = -1.84$ in Fluent and -1.720 in XFLR5. Velocity acceleration reaches 70.0 m/s ($1.40 U_\infty$). The adverse pressure gradient on the upper surface shifts natural transition in XFLR5 upstream to $x/c = 0.426$.



(a) Velocity magnitude contour ($V_{\max} = 70.0 \text{ m/s}$)

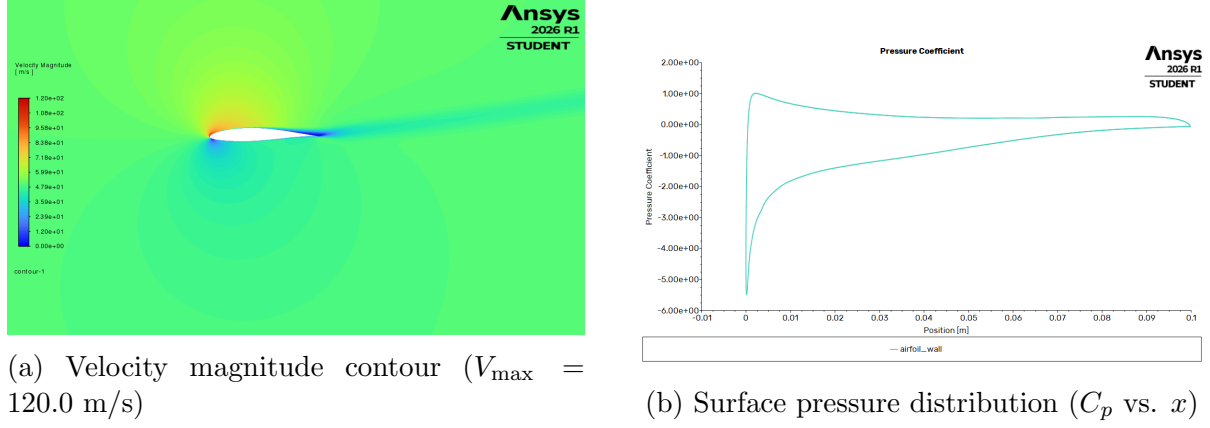


(b) Surface pressure distribution (C_p vs. x)

Figure 2: Aerodynamic flow field at $\alpha = 5^\circ$.

3.2.3 Pre-Stall Regime ($\alpha = 10^\circ$)

At $\alpha = 10^\circ$, near-peak lift is attained ($C_L = 1.2287$ Fluent vs. 1.1954 XFLR5, a $+2.79\%$ difference). An intense leading-edge suction spike develops ($C_{p,\min} = -5.51$ in Fluent, -4.446 in XFLR5), driving localized peak velocities up to 120.0 m/s ($2.40 U_\infty$, $M_{\text{local}} \approx 0.35$). The adverse pressure gradient forces boundary-layer transition upstream to $x/c = 0.0495$ (4.95% chord), bringing the boundary layers in Fluent and XFLR5 to near-identical turbulent extents. Wake shear layers visibly thicken, increasing drag to $C_D = 0.0352$.

Figure 3: Aerodynamic flow field at $\alpha = 10^\circ$.

4 Stall Mechanics and Numerical Unsteadiness at $\alpha = 15^\circ$

4.1 Physical Separation and Wake Dynamics

At $\alpha = 15^\circ$, the NACA 2412 enters aerodynamic stall. The velocity field indicates massive boundary-layer separation originating near the leading-edge suction peak and spanning the entire upper surface. The velocity contour illustrates a low-speed recirculation zone and strong wake deficit widening into a shear-layer wake.

4.2 Limit-Cycle Oscillations in Steady Solvers

When simulated using a steady-state formulation ($\partial/\partial t = 0$), the force monitors do not converge to a single static value. Instead, they settle into a regular, self-sustained sinusoidal limit cycle:

- **Drag Coefficient Bounds:** $C_D \in [0.105, 0.350]$ with an arithmetic mean $C_{D,\text{mean}} \approx 0.228$.
- **Lift Coefficient Bounds:** $C_L \in [0.360, 1.540]$ with an arithmetic mean $C_{L,\text{mean}} \approx 0.950$.
- **Periodicity:** Oscillations display a consistent period of approximately 23 iterations.

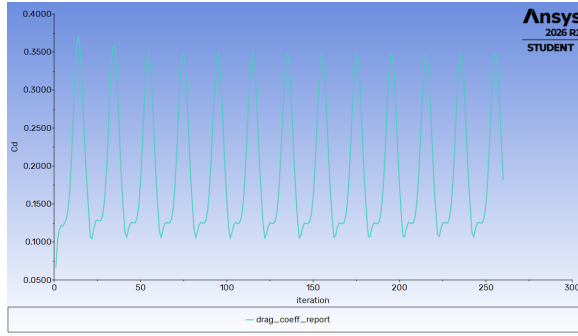
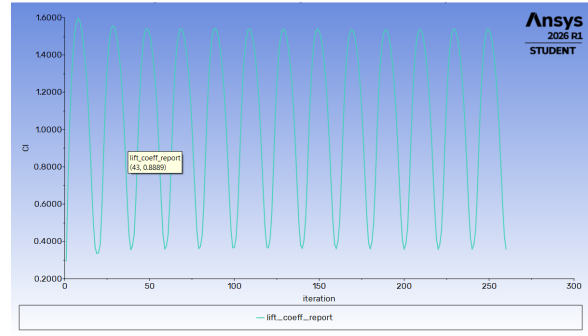
(a) Sustained limit cycle in C_D monitor(b) Sustained limit cycle in C_L monitor

Figure 4: Iterative monitor histories demonstrating physical unsteadiness and vortex shedding at $\alpha = 15^\circ$.

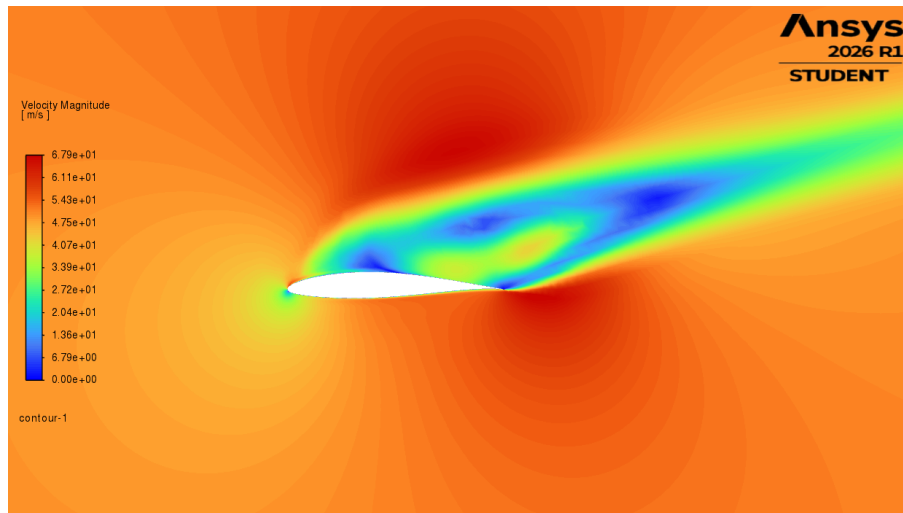


Figure 5: Velocity contour at $\alpha = 15^\circ$ displaying leading-edge boundary layer separation and a wide wake.

This behavior indicates that the flow physics has transitioned to an unsteady von Kármán vortex-shedding regime. The steady-state solver continuously cycles as vortices form, detach, and shed into the wake. Accurately resolving this regime requires a transient (time-accurate) formulation with time-step resolution $\Delta t \approx 2 \times 10^{-4}$ s (20–30 steps per convective timescale).

5 Conclusions

1. **Linear Regime Validation:** The $k-\omega$ SST model accurately predicts lift across the attached linear regime ($\alpha = 0^\circ$ to 10°), matching XFLR5 within 1.8% to 2.8%.
2. **Pressure Peak Resolution:** Numerical C_p profiles show consistent agreement with panel method distributions, properly resolving suction acceleration peaks up to $C_{p,\min} = -5.51$ at $\alpha = 10^\circ$.

3. **Drag Discrepancy Insight:** Fluent's fully turbulent boundary layer model accounts for the higher zero-incidence drag compared to XFLR5's e^9 -based natural transition run.
4. **Stall Identification:** At $\alpha = 15^\circ$, steady-state solver oscillations provide a clear numerical signature of the onset of stall, driven by large-scale separation and time-dependent vortex shedding.